

Fixing Windows Crashes When Enabling VirtualMachinePlatform (for WSL2)

When enabling the **VirtualMachinePlatform** (required for WSL2) causes Windows to crash (black screen or BSOD) on reboot, the problem is usually due to outdated drivers, firmware, or conflicting settings. To fix this reliably, follow these steps:

- **Update Windows & Drivers First:** Before doing anything else, make sure Windows itself is fully updated and all hardware drivers (especially chipset and graphics) are up-to-date. Microsoft experts recommend installing all pending Windows updates and downloading the latest drivers from your PC or motherboard manufacturer ¹ ². In particular, update your **BIOS/UEFI** firmware and your CPU/GPU drivers. For example, one report showed a BSOD (`VIDEO_TDR_FAILURE` from `igdkmdn64.sys`) caused by an old Intel GPU driver when enabling virtualization ³. Updating the BIOS and the Intel graphics driver eliminated the crash.
- **Boot into Safe Mode & Disable VirtualMachinePlatform:** If your system is already crashing on boot, restart into **Safe Mode** and turn off the VirtualMachinePlatform feature. In Safe Mode, press **Win+R**, run `optionalfeatures.exe`, and uncheck **Virtual Machine Platform** ⁴. (This is described in Microsoft's support Q&A: Safe Mode lets you disable the feature without crashing ⁵.) After unchecking, reboot normally. This should restore Windows to a stable state.
- **Restart and Apply Updates:** With the feature disabled, reboot Windows normally. Then revisit **Windows Update** and install any new updates available ¹. Also update any remaining drivers (for network, storage, graphics, etc.). It's important that Windows is fully patched before re-enabling virtualization features ¹ ².
- **Re-enable VirtualMachinePlatform (and Hyper-V):** Once updated, go to **Turn Windows features on or off** again. **Check** (enable) both **Virtual Machine Platform** (and **Windows Subsystem for Linux** if not already checked) ⁶. (On Windows Pro/Enterprise you may also enable **Hyper-V**, though WSL2 itself requires only the lightweight VirtualMachinePlatform component.) Click OK and allow Windows to reboot. If prompted, restart the PC. After reboot, Windows should install the necessary components.
- **Repair Hypervisor Service:** If simply re-enabling still fails, completely uninstall and reinstall Hyper-V and VirtualMachinePlatform. Use the Windows Features dialog or PowerShell to uncheck **Hyper-V** and **Virtual Machine Platform**, reboot, then check them again ⁷ ⁸. Finally, open an elevated Command Prompt and run:

```
bcdedit /set hypervisorlaunchtype Auto
```

then reboot ⁹. This ensures the Hyper-V hypervisor is configured to launch. Many users report that a clean uninstall/reinstall of these features, plus setting `hypervisorlaunchtype` to Auto, fixes stubborn virtualization issues ¹⁰ ⁹.

- **Verify BIOS/UEFI Virtualization Settings:** Make sure your BIOS settings are correct. Enter your BIOS setup (commonly by pressing **Del**, **F2**, etc. at boot) and enable all CPU virtualization options: on Intel systems enable **Intel VT-x**, **Intel VT-d** (or Intel Virtualization Technology), and (if present) **Intel Platform Virtualization Technology (VT-d)**. On AMD systems enable **SVM Mode** (Secure Virtual Machine). Also ensure **Second Level Address Translation (SLAT)** is supported (modern CPUs all have this) ¹¹. One Microsoft support post explicitly advises checking the BIOS's virtualization settings if enabling VirtualMachinePlatform caused a crash ¹¹.
- **Adjust Integrated GPU/UMA Memory (if needed):** On some systems (especially AMD APUs or systems with shared GPU memory), enabling virtualization can conflict with the GPU memory reservation. For example, AMD users have fixed similar black-screen issues by setting their **UMA Frame Buffer Size** to **Auto** (or increasing it to 1 GB) in BIOS ¹². If you have a combined CPU/GPU (like AMD APU or Intel with integrated graphics), check your BIOS for graphics memory settings (sometimes called DVMT or UMA Frame Buffer) and try setting it to **Auto** or a larger value. For Intel systems, ensure your integrated GPU driver is up to date (see above); outdated Intel graphics drivers (`igdkmd64.sys`) have been known to crash when virtualization is enabled ³.
- **Disable Conflicting Windows Security Features:** Some Windows security features can conflict with Hyper-V. In particular, **Core Isolation – Memory Integrity** (in Windows Security) is known to interfere. If “Memory Integrity” is enabled, try turning it **OFF** under **Windows Security → Device Security → Core Isolation**. (Several users report that disabling Core Isolation resolved WSL2 stability issues.) Conversely, some rare cases required re-enabling it via Group Policy. The key is that core isolation's hypervisor-based protections can clash with VirtualMachinePlatform; experiment by toggling it off if crashes persist.
- **Final Steps and Verification:** After re-enabling VirtualMachinePlatform and adjusting BIOS, reboot your PC one more time. Run `wsl --set-default-version 2` (or try installing a WSL2 distro) to verify WSL2 now works. Your PC should boot normally without freezing. If it still fails, double-check that Hyper-V, VirtualMachinePlatform, and Windows Subsystem for Linux are all enabled, and that `hypervisorlaunchtype` is set to Auto ⁹.

By meticulously updating BIOS/drivers and cycling the VirtualMachinePlatform feature off and on (often via Safe Mode), this method has resolved crashes for many users. In summary, the solution is to **fully update your system, disable the feature in Safe Mode, reinstall it cleanly**, and **ensure BIOS virtualization** is correctly set ⁴ ⁷. This approach addresses the common causes of crashes when installing WSL2.

Sources: Official troubleshooting steps from Microsoft support were used (Safe Mode disable/enable, BIOS update, reinstall features) ¹³ ¹ ¹⁰ ⁹, along with community solutions (update drivers, adjust UMA memory) ¹² ³, to compile a comprehensive fix. These recommendations have worked for affected users in multiple reports.

1 4 5 6 11 13 Windows crash when Virtual Machine Platform was enabling - Microsoft Q&A

<https://learn.microsoft.com/en-us/answers/questions/3985493/windows-crash-when-virtual-machine-platform-was-en>

2 3 BSOD when trying enable Virtual Machine Platform for WSL2 - Microsoft Q&A

<https://learn.microsoft.com/en-us/answers/questions/4245079/bsod-when-trying-enable-virtual-machine-platform-f>

7 8 9 10 How to fix "Please enable the Virtual Machine Platform Windows feature and ensure virtualization is enabled in the BIOS." - Microsoft Q&A

<https://learn.microsoft.com/en-us/answers/questions/4039153/how-to-fix-please-enable-the-virtual-machine-platf>

12 Windows 11 crashes (black screen) after enabling the "Virtual Machine Platform" for WSL2 - Super User

<https://superuser.com/questions/1703165/windows-11-crashes-black-screen-after-enabling-the-virtual-machine-platform>